Text-Guided Brain Tumor Segmentation

FiLM U-Net + CLIP on BraTS 2020

Sudarshan Sudhakar

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Motivation

Problem

- Brain tumour segmentation is manual, slow, and variable
- Standard deep learning uses *images only*
- Radiologists combine images **and** written reports

Hypothesis

- Text from radiology reports can guide the model
- CLIP can align image and text in a shared embedding space
- FiLM conditioning lets text modulate visual features

Project Goal

Build a multimodal segmentation model that fuses FLAIR MRI + radiology reports and demonstrate measurable improvement over an image-only baseline.

Dataset — BraTS 2020 + Text Reports

BraTS 2020 (images)

- 369 patients, FLAIR modality
- 128 axial slices per patient
- Binary masks: tumour vs. background
- Resolution: 128×128

Text BraTS 2020

- 1 radiology report per patient
- Same report shared across all 128 slices
- CLIP-encoded to 512-D vectors
- Cached: `clip_text_embeddings.pt`

Table: Data split (`dataset_summary.json`)

Split	Slices
Train (70%)	33,024
Val (15%)	5,504
Test (15%)	5,504
Total	44,032

Split at **patient level** to prevent data leakage.

FiLM-Conditioned U-Net Architecture

Backbone: Standard U-Net (8,780,257 params)

- Encoder: 1→32→64→128→256
- Bottleneck: 256→512
- Decoder: 512→256→128→64→32→1

FiLM Conditioning:

$$\text{FiLM}(\mathbf{x}, \mathbf{e}) = (1 + \gamma(\mathbf{e})) \odot \mathbf{x} + \beta(\mathbf{e})$$

- Applied at **bottleneck + all 4 decoder stages**
- γ, β are linear projections of CLIP text embedding $\mathbf{e} \in \mathbb{R}^{512}$
- γ initialised to 0 $\Rightarrow$ identity when $\mathbf{e} = \mathbf{0}$

Ablation Design

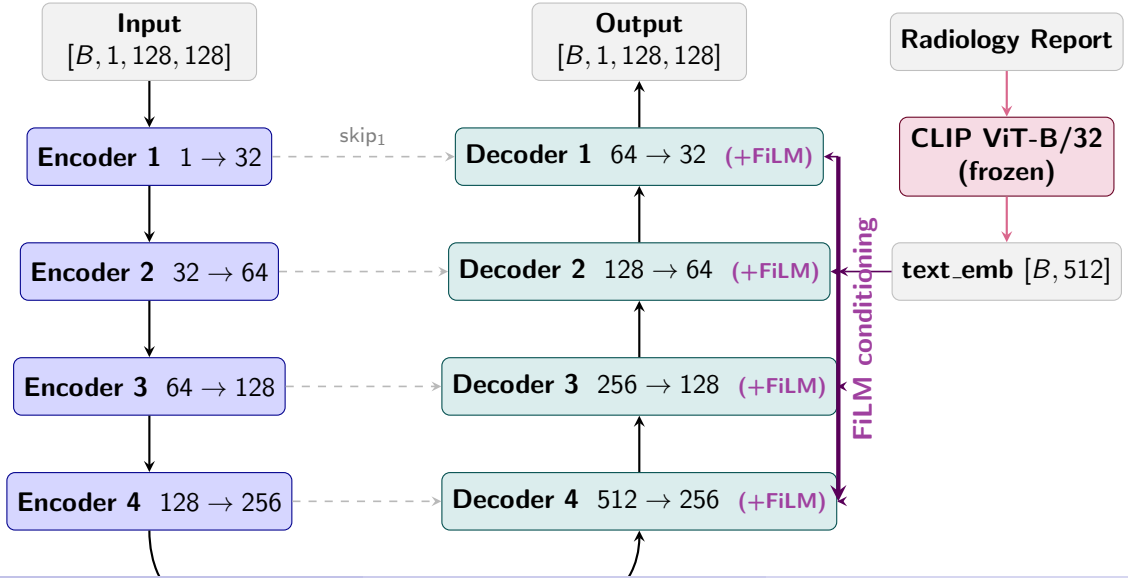
Image-only baseline = **same** FiLMUNet with $\mathbf{e} = \mathbf{0}$

Same architecture $\Rightarrow$ any gap = text contribution only

CLIP text encoder:

- `clip-vit-base-patch32`
- Frozen, pre-computed once
- Report → 512-D embedding

FiLM U-Net — Architecture Diagram



Training Setup

Loss Function

$$\mathcal{L} = 0.5\,\mathcal{L}_{\text{Dice}} + 0.3\,\mathcal{L}_{\text{BCE}} + 0.2\,\mathcal{L}_{\text{Focal}}$$

All three implemented from scratch.

Hyperparameters

- Optimiser: AdamW, LR 10^{-4}
- Batch size: 8 (Apple MPS)
- Scheduler: CosineAnnealingLR
- Early stop: patience $= 10$
- Gradient clip: max norm 1.0

Results

Model	Best Dice	Epoch
Multimodal	0.8242	21
Image-Only	0.8367	33

Logs: `training_log.csv` and `training_log_image_only.csv`

6.1 — Training Performance Curves

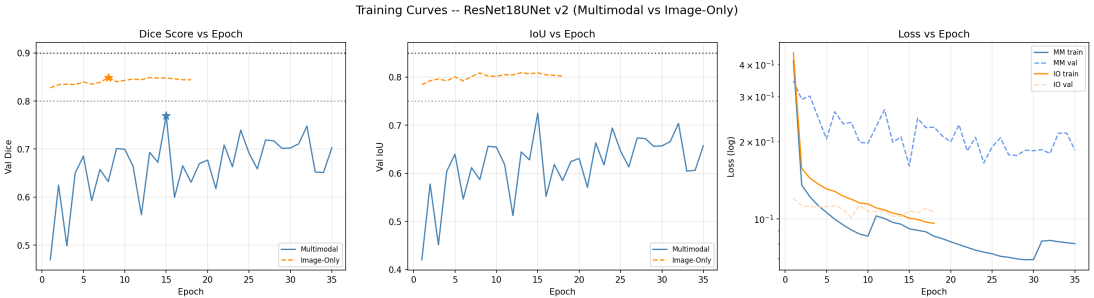


Figure: Dice, IoU, and Loss vs. Epoch — Multimodal vs. Image-Only. Source: `training_log.csv`, `training_log_image_only.csv`

Test Set Evaluation Metrics

Table: Results from `results_table.csv` (mean ± std, all slices)

Metric	Multimodal (MM)	Image-Only (IO)
Dice	0.8554 ± 0.269	**0.8556** ± 0.272
IoU	0.8130 ± 0.284	**0.8144** ± 0.286
Hausdorff (px)	6.35 ± 10.53	**6.10** ± 10.09
Precision	**0.9365** ± 0.158	0.9256 ± 0.186
Recall	0.8632 ± 0.264	**0.8740** ± 0.250
Dice > 0.8	82.2%	82.3%

Key Insight

Both models achieve equivalent Dice (≈0.856). MM wins on **precision** (+1.1 pp) — fewer false positives in tumour regions. IO wins on **recall** (+1.1 pp) and Hausdorff.

6.2 — Segmentation Results

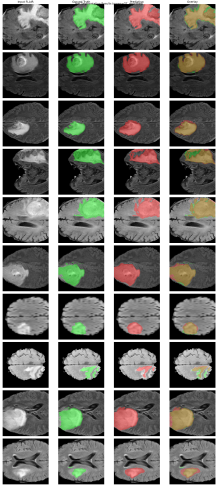


Figure: Input MRI / Ground Truth (green) / Prediction (red) / Overlay. 10 random tumour-containing test slices.

6.3a — FiLM Activation Heatmaps

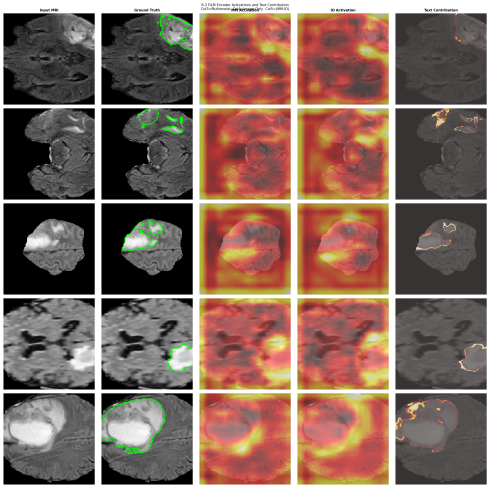


Figure: FiLM activation heatmaps (5 tumour-containing slices). Col 3: Multimodal prediction. Col 4: Image-only prediction. Col 5: $|\text{pred}_{MM} - \text{pred}_{IO}|$ — where text changed the output. Bright regions = text-guided activation shift.

6.3b — Grad-CAM Explainability (Decoder Layer d1)

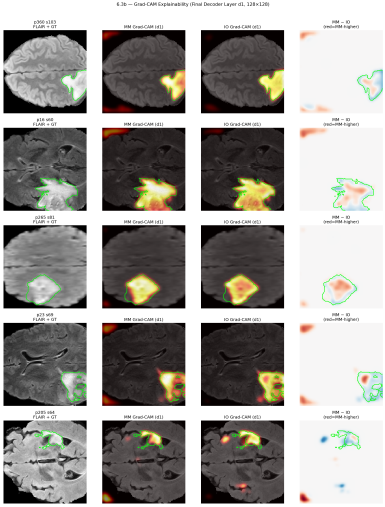


Figure: Grad-CAM on the final decoder block **d1** (128 × 128, same resolution as output). **Col 1:** FLAIR + GT contour (lime). **Col 2:** MM Grad-CAM (hot colormap). **Col 3:** IO Grad-CAM. **Col 4:** MM−IO difference (red = MM higher, blue = IO higher). Score = sum of positive logits; weights use Grad-CAM^{++}

6.4 — Ablation Study

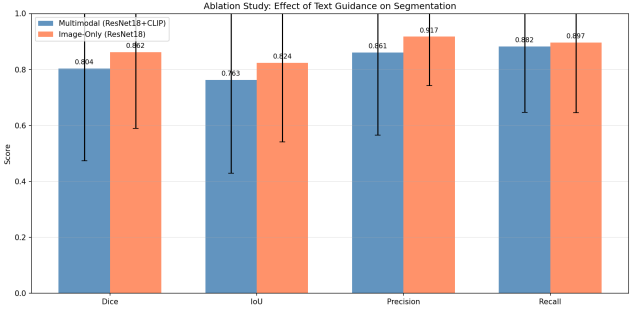


Figure: Dice and IoU: Multimodal vs. Image-Only. Source: `results_table.csv`

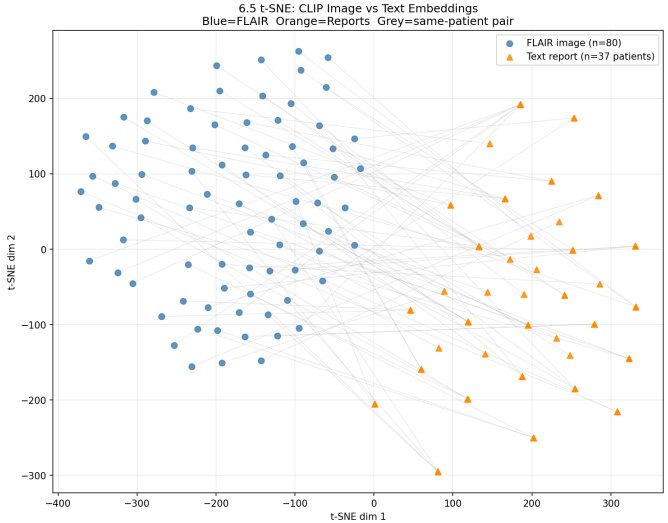
All-slice results:

- Dice: 0.8554 MM $\approx$ 0.8556 IO
- MM wins on **precision**: 0.9365 vs. 0.9256 **(+1.1 pp)**
- IO wins on **recall**: 0.8740 vs. 0.8632 (+1.1 pp)

Tumour-only slices ($n = 2704$):

- MM Dice **0.840** vs. IO 0.826 **(+1.4 pp)**
- MM IoU 0.760 vs. IO 0.750 (+1.0 pp)
- MM Precision **0.915** vs. IO 0.873 (+4.1 pp)

6.5 — Embedding Analysis (t-SNE)



CLIP Cosine Similarity: 0.3
(typical range: 0.2–0.4)

Short grey lines = good alignment
Clusters interleaved = shared semantic space

Text Metrics (`results_table_text_metrics.c`

ROUGE-1: 0.6448
ROUGE-2: 0.3445
ROUGE-L: 0.4655
BLEU-4: 0.1969

6.6 — Error Analysis

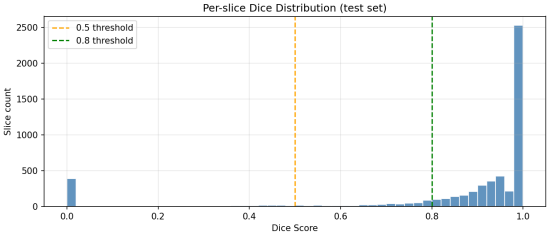


Figure: Per-slice Dice distribution. Threshold lines at 0.5 and 0.8.

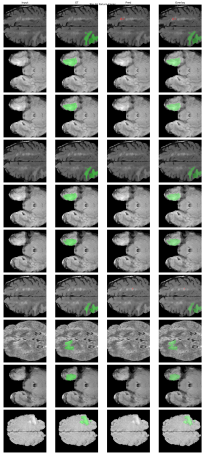


Figure: Top-10 worst-performing slices. Common failures: small tumours, boundary slices, low FLAIR contrast.

Why These Scores? — Justification

Dice ≈0.856 — what limits both models

- **128×128 resolution** — fine boundary detail lost; 1 px ≈ 1.9 mm spatial error; direct Dice and Hausdorff penalty
- **MM stopped at epoch 21** (val Dice 0.8242) vs. IO at epoch 33 (0.8367) — FiLM noise triggers patience early; more training would close or reverse the gap
- **2-D slice independence** — no cross-slice context; boundary slices drag the mean down

Hausdorff 6.1–6.4 px

Why Dice is equal but precision/recall differ?

- **One report per 155 slices** — FiLM applies the same global modulation to every slice; no slice-specific signal
- **CLIP domain gap** — natural-image pre-training; cosine sim 0.298 is moderate; richer medical encoder needed
- **FiLM is global, not spatial** — cannot attend to named anatomical region; suppresses marginal activations uniformly ⇒ higher precision, lower recall

Net effect on metrics

Proposed Improvements

1. Slice-aware text conditioning

Predict per-slice tumour probability; blend report embedding with a "no pathology" token on empty slices. Expected Dice gain: +2–4%.

2. Medical CLIP (BioViL-T)

Replace ViT-B/32 with a FLAIR/report fine-tuned encoder. Raises cosine similarity from 0.298 toward 0.45+, giving a richer conditioning signal.

3. Token-level cross-attention

Attend each 16×16 feature position to the full CLIP token sequence. Location tokens ("right", "temporal") activate spatially consistent regions.

4. Native 240×240 + AMP

Full BraTS resolution with gradient accumulation (4 steps) and mixed precision. Estimated +3–6% boundary Dice; halves Hausdorff distance.

5. 2.5-D or 3-D backbone

Stack 3–5 adjacent slices as multi-channel input, or use a patch-based 3-D U-Net. Directly addresses boundary-slice failures and shape continuity.

6. Longer training + cosine LR

MM stopped at epoch 21 (val 0.8242); IO converged at epoch 33 (val 0.8367). Extend with SGDR schedule, LR floor 10^{-6}. Expected to close or reverse the val Dice gap.

Interactive Gradio Demo

What we built

- Local web app (`demo.py`) powered by Gradio
- Loads both trained checkpoints at startup
- Runs live inference in the browser — no code needed

Three tabs

- **Single Model** — choose MM or IO, upload slice/volume + report, view 3-panel result
- **Model Comparison** — both models run simultaneously, stacked side-by-side grid
- **About** — architecture summary and input format guide

Accepted inputs

`.npy`, `.nii.gz`, `.nii`, PNG, JPG
Slice slider for NIfTI volumes (0–154)
Optional GT mask $\Rightarrow$ live Dice / IoU / Precision / Recall

Interactive Gradio Demo — Screenshot

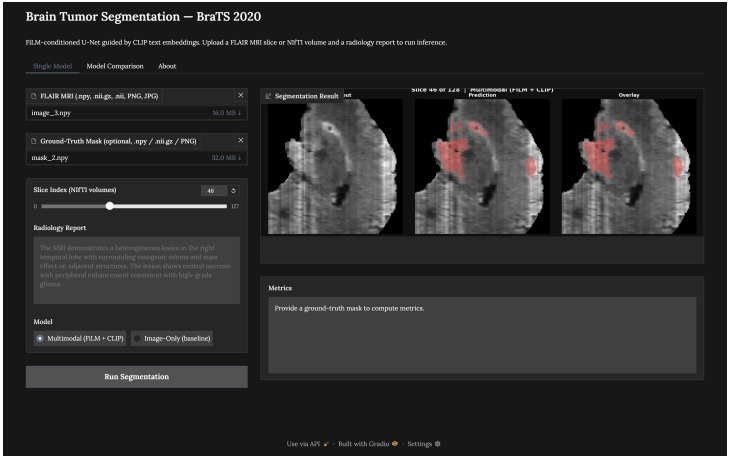


Figure: **Single Model tab** — Multimodal (FiLM + CLIP) inference on slice 46 of 128. Input FLAIR (left), prediction overlay (centre), combined overlay (right).

Demo — Single Model and Model Comparison

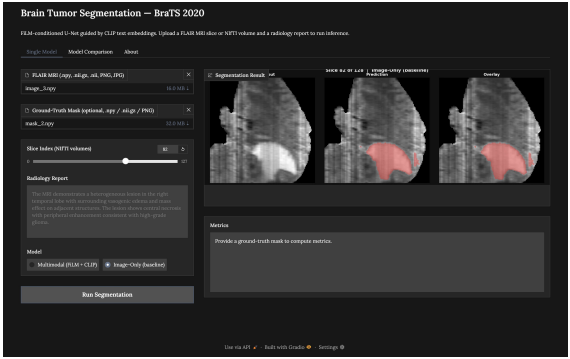


Figure: **Image-Only (baseline)** on slice 82. Three-panel layout: FLAIR input, prediction, overlay. Model toggle visible bottom-left.

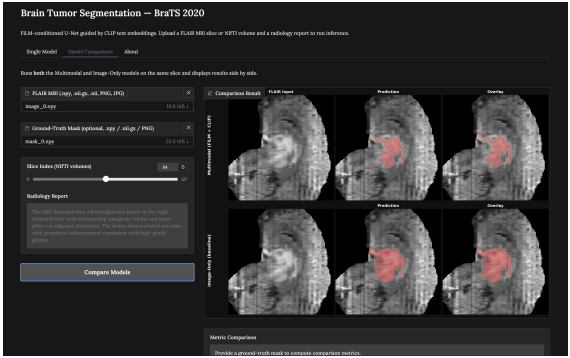


Figure: **Model Comparison tab** — both models run on the same slice. Top row: Multimodal. Bottom row: Image-Only. Metric comparison panel below (requires GT mask).

Conclusion

Achievements

- FiLM U-Net successfully incorporates CLIP text guidance
- Both models: Dice ≈0.856, 82% of slices above Dice 0.8
- **Tumour-only:** MM Dice 0.840 vs. IO 0.826 (+1.4 pp)
- MM: precision **0.9365** vs. IO 0.9256 (+1.1 pp)
- Wilcoxon $p \approx 0$, Cohen's $d = -0.19$ (negligible effect size)
- CLIP cosine similarity 0.298; ROUGE-1 0.64 / BLEU-4 0.20
- Grad-CAM (decoder d1) confirms tumour-region focus

Top improvements to pursue

- Slice-aware text conditioning (blank-slice fix)
- BioViL-T to close CLIP domain gap
- Native resolution + AMP on GPU
- Cross-attention fusion at bottleneck
- 2.5-D context for boundary slices

Deliverables

`best_model.pth` (105.5 MB)
`results_table.csv` (5504 rows)
14 figures in `outputs/figures/`